Portfolio · rubenayla.xyz
linkedin.com/in/rubenayla
+34 674 884 126
Griñón, Madrid (open to relocate)
ruben.jimenezmejias@gmail.com

SKILLS

- Autonomy software (Python, ROS2)
- Computer vision (YOLO, OpenCV)
- Edge AI deployment (Jetson, TensorRT)
- Embedded C (STM32, ESP32, PIC18)
- PCB design (KiCad, Altium, Eagle)
- Closed-loop control (PID / PD)
- Sensors & perception (stereo vision, Hall)
- CAN · UART · I²C
- SolidWorks · Fusion 360
- System integration

LANGUAGES

- **Spanish**: native
- **English**: C1 (professional, daily use)
- **German**: beginner (A1, learning)

SELECTED PROJECTS

- Cyberwheel: open-source electric unicycle, hardware and firmware
  github.com/rubenayla/cyberwheel
- Partle: local-stores product search
  github.com/rubenayla/partle
- Invest: stock-valuation engine (ML + fundamental models)
  github.com/rubenayla/invest

HOBBIES

- Saxophone in my town's wind band
- Teaching piano lessons
- Riding electric unicycles

OTHER

- EU citizen (Spanish), no sponsorship required
- Moving at my own expense, no relocation package needed
- Driving license (B)

RUBEN JIMENEZ MEJIAS

Hands-on robotics engineer who takes robots from nothing to working in the physical world. I cover the whole stack, from the high-level code that makes a vehicle drive itself down to the firmware and the PCBs it runs on. Aerospace Engineering background.

EXPERIENCE

Chief Engineer, Driverless Section — Ü Motorsport

2024 – present

- **Led the go-kart from nothing to a working self-driving vehicle**: perception and control software (Python, ROS2, NVIDIA Jetson Orin), embedded firmware, and the steering it drives. YOLOv11 perception at ~80 FPS (TensorRT FP16). Documented at *github.com/UM-Driverless/kart-docs*.
- Designed the kart's electric steering: custom 3D-printed planetary reducer (Fusion 360, nylon + metal sun gear) driven by a closed-loop PID controller on ESP32 with Hall-sensor angle feedback (PWM).
- Ran the section: personnel, budgets, timelines, sponsorships.

Mechatronics Engineer — Ü Motorsport · Formula Student

2020 – 2024

- Built the cone-detection perception and driving pipeline for the Formula Student driverless car (Python). Re-architected it to run multithreaded, taking perception from ~0.5 to ~40 FPS. Top contributor (237 commits, ~5× the next): *github.com/UM-Driverless/driverless*. Explainer: *youtu.be/wZSFr2eYE4M*.
- Full-vehicle hardware: electronics and wiring, CAD parts for laser cutting, PCB design (Eagle), PIC18 firmware, and CAN / UART / I²C. Designed the car's electric steering (torque calculations, SolidWorks); secured new sponsors.

Software Engineering Intern — Radian Space

Remote · Feb 2025 – Oct 2025

- Built mechanisms in thermal-modeling software for satellite systems (Python): reference-frame handling, geometry data, and code validation.

Research Developer — URJC (Next-Generation EU funds audit)

Jan 2024 – Mar 2025

- Built a web-scraping pipeline for news and public tenders; managed scraping servers and fine-tuned a binary classification model for fraud signals.

Aeromodelling Team Member — COSMOS

2018 – 2020

- CAD design and early physical testing of a VTOL drone, among other aeromodelling projects.

EDUCATION

Bachelor's Degree in Aerospace Engineering (Aerospace Vehicles)

Universidad Rey Juan Carlos, Madrid · 2017 – 2026 · Graduated, 8.8/10

Professional Grade, Piano

Rodolfo Halffter Conservatory, Móstoles · 2010 – 2020